

Local Business Lead Machine

An n8n pipeline that scrapes Google Maps, audits each business's website, scores every lead, and drafts a personalized pitch — with outreach only ever created as a Gmail draft after human approval.

n8n

Apify

Firecrawl

OpenAI

Google Sheets

1 Executive Summary

Cold outreach usually starts with hours of manual research: finding businesses, checking their websites, guessing what's missing, and writing a pitch from scratch. This workflow automates the entire research phase. One form submission scrapes Google Maps, audits every business's website for concrete gaps, scores each lead 0–100 on transparent rules, and — only for leads worth pursuing — has AI write a grounded pitch referencing the gaps actually found. Nothing is emailed until a human approves it.

2 Key Metrics

Trigger	A form: business type, city, country, max results, service to pitch
Lead Source	Apify's Google Places crawler, deduplicated by place and website domain
Website Audit	Direct HTTP fetch first; Firecrawl as a fallback for sites that fail or hide their email
Scoring	Transparent rule-based score, 0–100, with no AI involved in the scoring itself
AI Pitch Writing	OpenAI writes subject, body and a call opener — Hot/Warm leads only
Outreach Safety	A separate scheduled flow only processes Approved leads, and only ever creates Gmail drafts
Estimated Cost	\$0 — Apify's free credit, Firecrawl's free tier, and OpenAI via Gateway credits

0–100

LEAD SCORE RANGE

20

DAILY OUTREACH CAP

\$0

ESTIMATED RUNNING COST

3 Business Value

- ✓ **Hours of manual prospecting replaced** — one form submission — business type, city and service in, a scored lead list out.
- ✓ **Pitches are grounded in reality** — the AI only references gaps the workflow actually detected on the real website.
- ✓ **AI spend stays under control** — cold leads are saved but skipped for pitch-writing entirely, so API calls stay proportional.
- ✓ **Full control before anything goes out** — outreach only fires for leads marked Approved, and even then it only drafts.

4 Architecture Overview

Two independent flows share one Google Sheets database, plus a dedicated error-alert branch.

FLOW	TRIGGER	PURPOSE
Lead Discovery & Audit	Form submission, on demand	Scrapes Google Maps, audits websites, scores leads, and drafts pitches for the strongest prospects
Approved Outreach	Scheduled, weekdays 10 AM	Turns leads marked Approved into Gmail drafts, up to a daily cap
Error Alerts	n8n Error Trigger, on failure	Posts a Slack alert naming the failed node and execution

5 How It Works

Lead Discovery & Audit	Approved Outreach
1 A form collects business type, city, country, max results and the service being pitched. 2 Apify's Google Places crawler returns matching businesses; results are deduplicated by domain. 3 Each website is fetched directly, with Firecrawl as a fallback for sites that fail or hide their email. 4 A transparent 0–100 rule-based score sorts each lead into Hot, Warm or Cold. 5 For Hot/Warm leads only, OpenAI drafts a subject, body and call opener grounded in the detected gaps.	1 A weekday 10 AM trigger reads the Leads sheet and filters to Approved leads with a usable email. 2 Only the first 20 eligible leads in the run are processed, capping the daily volume. 3 A Gmail draft is created per lead using the AI-written subject and body — never sent automatically. 4 Each processed lead is marked Gmail Draft with a timestamp, and Slack reports the batch results.

6 Reliability & Security Design

- ◆ **AI never invents facts** — the pitch prompt explicitly forbids fake statistics, invented names, or gaps that weren't detected.
- ◆ **Cold leads cost nothing extra** — leads below the Warm threshold are saved but routed around the AI pitch step.
- ◆ **Human approval gate before any outreach** — the outreach flow only ever processes leads a person has marked Approved.
- ◆ **Draft-only sending** — even approved leads only get a Gmail draft created, never an automatic send.
- ◆ **Resilient scraping** — website fetch failures continue the workflow rather than stopping it.
- ◆ **Rate-limited and capped** — fetches, AI calls and outreach are all batched, delayed and capped at safe limits.

7 Setup Checklist

- ✓ Add an Apify credential (Header Auth) — the workflow uses the compass/crawler-google-places actor.
- ✓ Add a separate Firecrawl credential (Header Auth) for the website audit fallback.
- ✓ Point every Google Sheets node at your spreadsheet, with a Leads tab and a Runs tab.
- ✓ Connect Slack and Gmail credentials, and choose your alerts channel.
- ✓ Start with Max Results = 20 on your first few searches to conserve Apify credits.

8 Node-by-Node Map

Search & Scrape	Start Lead Search → Build Apify Input → Apify Scrape Google Maps → Normalize & Dedupe
Website Audit	Fetch Website → Detect Signals → Firecrawl Fallback → Merge Website Results
Scoring & Pitch	Opportunity Score → Needs Pitch? → Write Pitch (AI) → Attach Pitch
Approved Outreach	Weekdays 10AM → Get Approved Leads → Daily Cap (20) → Create Gmail Draft → Mark as Drafted

SUMMARY

Designed and built end-to-end in n8n, demonstrating multi-source lead scraping, rule-based opportunity scoring, grounded AI copywriting, and a human-approval-gated outreach loop.